

Empiricist

12+3 · course smoke — NOT VisionMix / ViT-G / Franka

E1 L_ctx + E2 $\lambda \in \{0, 0.5, 1\}$ on tiny clips. Do NOT claim 0.307 / 7.71 / Franka 80% / ViT-G.
Temporal mask = GAP (extension only).

Shared spine — tattoo every deck

Six claims that must land in every seat

1 Unlock dense features in the JEPA world-model line — L_ctx + deep SS + VisionMix/ViT-G.

2 CLEAR lineage: I-JEPA → V-JEPA → V-JEPA 2 → V-JEPA 2.1 → 2-AC.

3 Franka grasp 60→80%; latent nav ~10× cheaper than NWM.

4 Triangle: SigLIP/DINO | MAE | JEPA WM (now dense).

5 Must: NYU 0.307; ADE 22.2→47.9; Ego4D 7.71; Tables 6-7; L_ctx alone hurts globals / deep SS restores.

6 SSv2 77.7; EK100 40.8; Tartan ATE 5.687 / 10.6s vs NWM 103.2s.

Table 1 — recipe stack (cite; smoke direction)

Authoritative ablation extract

Stage	IN1K	SSv2	NYU↓	ADE
V-JEPA 2	82.2	72.8	0.682	22.2
+ Context Loss	72.6	62.5	0.474	33.8
+ Multi-level	80.8	72.1	0.463	38.6
+ VisionMix	81.6	72.6	0.418	40.8
+ Multimodal tok	81.6	72.6	0.415	41.4
+ Scale	84.8	76.1	0.365	47.1
+ Cool-down	85.5	77.7	0.307	47.9

Table 2 — λ weighting (course grid)

Weighted + warmup $\lambda=0.5$ chosen + final $\lambda=0.5$ video / 0.7 image

Setting	ADE	SSv2	Note
$\lambda=0$ (VJ2)	22.2	72.8	baseline
$\lambda=0.2$ cte	29.6	62.5	dense $\uparrow$ global $\downarrow$
$\lambda=0.5$ cte	27.5	53.8	worse trade
$\lambda=1.0$ cte	24.6	51.1	overweight
$\lambda=0.5$ + warmup	32.2	61.5	stabilizes
$\lambda=0.5$ weighted+wu	33.8	62.5	CHOSEN

Architecture — $L_{\text{dense}} = L_{\text{predict}} + L_{\text{ctx}}$

Fig. 4 · deep SS 4 levels · multimodal 2D/3D tok

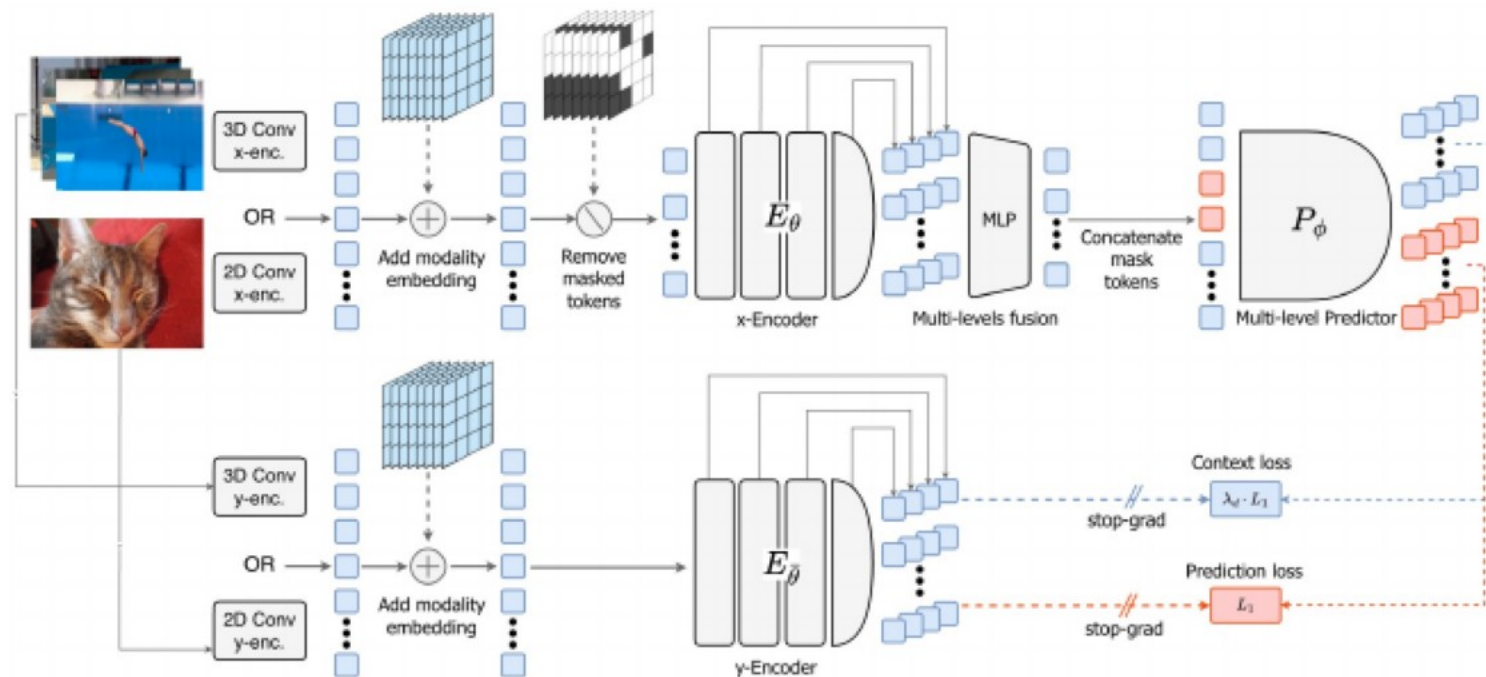


Figure 4 V-JEPA 2.1 Detailed Architecture. Images and videos are processed by respectively either a 2D or 3D Convolutional patch embedding. Then, 3D Rotational Positional Encoding (RoPE) and learnable modality embedding are added. The x -encoder processes the visible tokens and outputs multi-level embeddings formed by concatenating the normalized output from intermediate encoder blocks. Then, a MLP fuses this multi-level representation and reduces its dimensionality. These context tokens are concatenated with learnable mask tokens carrying spatio-temporal positional information. The predictor processes the combined sequence and produces multi-level predictions for the masked tokens. Training uses two different losses: (i) an L1 loss on masked-token predictions (the original V-JEPA objective), and (ii) a distance-weighted L1 loss on nearby context tokens, both supervised using the y -encoder multi-level outputs.

Unit-test: mask indices; L_{ctx} only on C; multi-level pred shapes; finite L1; EMA.

Course smoke contract — Y / Partial / N

Ablation Bot authoritative · no jobs launched this quiet run

Y · P0

E1 L_ctx on/off
E2 $\lambda \in \{0, 0.5, 1\}$
tiny video/frame proxy

Partial · P0

E3 deep-SS lite
1 vs 4 levels
fixed tiny data

Partial

E4/E7 frozen probes
E6 multimodal tok
if budget

N / stub

VisionMix · ViT-G
full Franka T6
nav vs NWM

GAP

Temporal mask
NOT paper ablation
label EXTENSION

Honesty

Never claim 0.307
7.71 / 80% / ViT-G
from smoke

Hypotheses — all seats consume this pack

Falsifiable directions on tiny clips

H-dense Weighted L_ctx improves dense proxy vs masked-only — mirrors T1 direction.

H-tradeoff Without deep SS, L_ctx hurts global clip classification — falsifiable T1/T2 tension.

H-deep Multi-level supervision recovers global accuracy while keeping most dense gains.

H-temporal EXTENSION ONLY — stronger temporal tubes may help motion probes; NOT paper ablation.

H-prior Frozen 2.1 beats 2 on dense probes when ckpts exist; smoke twins show same SIGN.

H-transfer Better dense → lower latent goal-distance / proxy grasp on sim — NOT Franka 80%.

GPU honesty — ruthless video smoke

Paper ViT-G / Franka = cite only

E1 L_ctx on/off ViT-S	1×24-40GB	12-36 h	Y
E2 $\lambda \in \{0, 0.5, 1\}$	same	+0.5-1× E1	Y
E3 deep-SS lite	same+20%	+0.5-1×	Partial
E7 frozen probe	1×16-24GB	1-6 h	Y
VisionMix / ViT-G	multi-node	days-week	N
Franka CEM	lab + A100	lab schedule	N

Temporal mask = GAP

Do not claim paper replication

NO temporal-masking ablation in-paper

Temporal Mask Scale fixed [1.0, 1.0] (Table 12) — same strategy as V-JEPA 2. Predictor 24 vs 12 stated but not singly ablated. Any course tube-mask sweep = EXTENSION only (prefer skip). Link [ijepa-quiet H-mask](#) as analogy only.

Robot = Table 6–7 cite only or Partial stub (frozen-enc / sim). Never claim Franka 80% from course.

Success bar — local vs NOT expected

Pass = paper-aligned TRENDS + honesty logged

L_ctx unlocks dense

OK: dense proxy $\uparrow$ vs masked-only

NOT: ADE 33.8 / NYU 0.474 abs

Dense hurts globals

OK: global recovers with deep SS

NOT: match 72.8 SSv2

Weighted λ best

OK: weighted $\geq$ fixed mid

NOT: full λ schedule fidelity

Beats VJ2 dense

OK: same-sign gap on mini probes

NOT: 0.307 / 47.9 abs

+20% grasp

OK: proxy $\uparrow$ OR document blocker

NOT: Franka 80% / CEM wall

10x nav

OK: cite only / toy embedding CEM

NOT: 5.687 ATE / 10.6s

Numbers tattoo — pin these

Reconciled with KANTA_PACK · shared across seats

0.307

NYUv2 RMSE linear
beats DINOv3-7B 0.309

22.2→47.9

ADE path T1
33.8→38.6→47.9

7.71

Ego4D STA mAP
ViT-G

40.8

EK100 Action R@5
ViT-G

77.7%

SSv2 attentive
72.8→62.5→72.1→77.7

60/70/80

Franka grasp n=10
matched→H=8 CEM

5.687

Tartan Drive ATE
10.6s vs NWM 103.2s

~10×

latent nav cheaper
than NWM SD-VAE

L_ctx

alone hurts globals
deep SS restores

33.8/62.5

T2 weighted $\lambda=0.5$
ADE / SSv2 trade-off

ViT-G 2B

VisionMix163M
+ cool-down 64@384

distill B/L

ViT-L $\approx$ G on
many probes (T11)

Empiricist — close

Takeaways + exam bite

1. Cite Table 1 recipe + Table 2 λ ; architecture $L_{\text{dense}}=L_{\text{predict}}+L_{\text{ctx}}$; deep SS 4 levels.
2. Course: E1+E2 Y on tiny clips; E3 Partial; never claim paper absolute %.
3. Temporal mask = GAP — any sweep is extension, not replication.
4. Robot = Table 6-7 cite or stub only.

EXAM BITE

Empiricist smoke validates L_{ctx} / λ / deep-SS directions on tiny clips; paper owns 0.307, 7.71, Franka 80%, ViT-G.

Exam bite — Empiricist

Honesty line

Run E1 L_ctx on/off and E2 $\lambda \in \{0, 0.5, 1\}$ on tiny short clips with ViT-Tiny/S. Expect dense proxy $\uparrow$ and global $\downarrow$ without deep SS. Do not claim NYU 0.307, Ego4D 7.71, Franka 80%, or ViT-G from course. Temporal-mask sweeps are extensions — the paper fixed [1.0, 1.0].